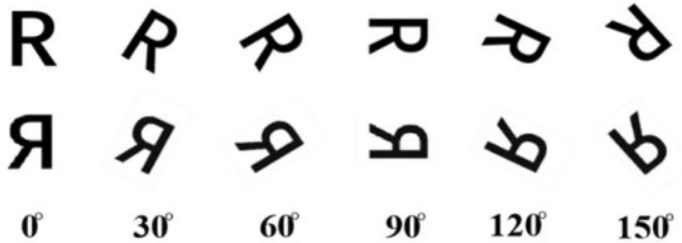
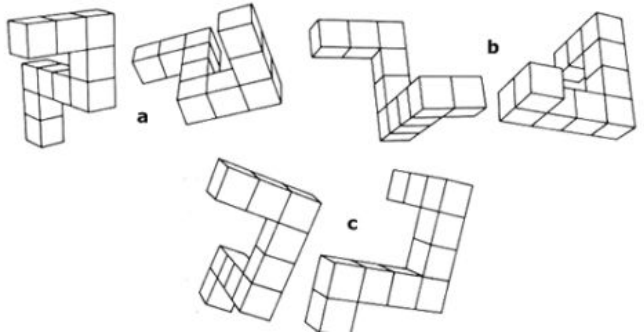
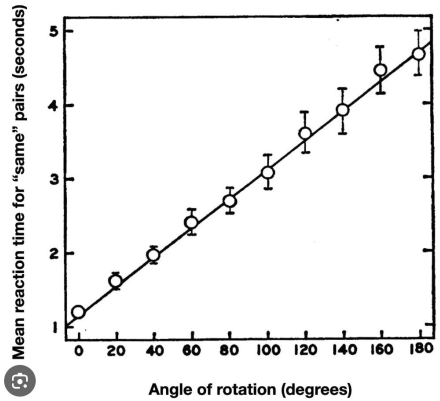
Mental Rotation

Shepard, R. N., & Metzler, J. (1971). Mental Rotation of Three-Dimensional Objects. *Science*, *171*(3972), 701–703. https://doi.org/10.1126/science.171.3972.701





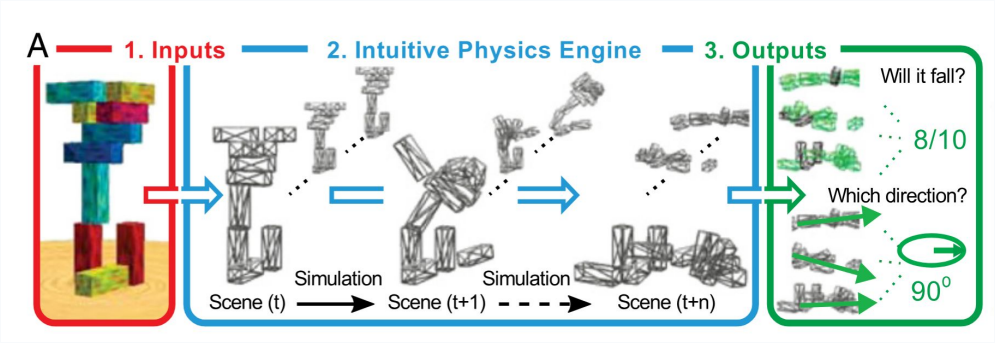


A "Physics Engine" in the Brain?

- Battaglia, Hamrick, & Tenenbaum (2013). Simulation as an engine of physical scene understanding. *PNAS*
- Fischer, Mikhael, Tenenbaum & Kanwisher (2016). Functional neuroanatomy of intuitive physical inference. *PNAS*
- Ullman, Spelke, Battaglia, Tenenbaum (2017). Mind Games: Game Engines as an Architecture for Intuitive Physics. *Trends in Cognitive Sciences*

Humans understand complex visual scenes by **running mental probabilistic simulations** (like a computer game's physics engine)

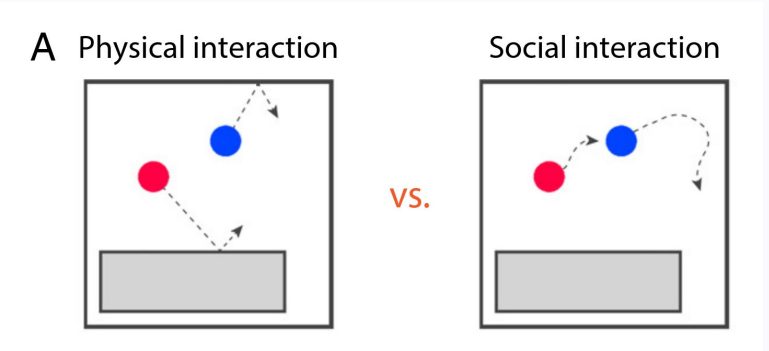
It allows them to predicts stability, collisions, and movement.



Neuroscientific evidence:

Intuitive physical inference engages a dedicated brain network,

separate from abstract reasoning, …

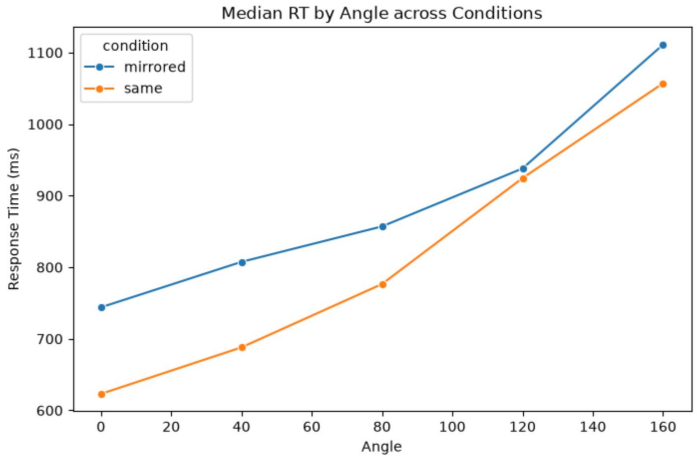


https://youtu.be/MI4PznE4AxE?si=c6KziN_5-YPkl6kO

Results from your own Data

16 of you sent me the csv result file after running
https://github.com/chrplr/goxpyriment/tree/main/examples/Mental-Rotation-2D

Descriptive statistics



Open the python notebook at https://github.com/chrplr/mental-rotation-analysis

1. Option one: In https://colab.research.google.com:

```
!git clone https://github.com/chrplr/mental-rotation-analysis
%cd mental-rotation-analysis
```

Then: File/open `MentalRotation2D-2026.ipynb`

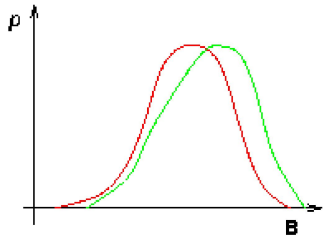
2. Option two: locally on your computer (you need Python and Git (https://git-scm.com/install))

```
pip install numpy jupyter pandas seaborn
git clone https://github.com/chrplr/mental-rotation-analysis
cd mental-rotation-analysis
jupyter notebook
```

Crash-course in inferential statistics

Objective: to compare two populations on a given continuous parameter (e.g. “is the average height of Germans larger than French ?”)

This is a question about a property of the **distributions** of heights in both populations.



Unless we have access to data for whole populations, we need to resort to **sampling**

Suppose we draw two random samples of French and German humans (N=4 in each group).

Here are the heights (in mm)

German	French
182	168
171	176
178	157
189	175
180	169

To determine whether these observations fit with the hypothesis of a real difference in average height between the populations, we would want to assess:

P(avg. height of male and female differs in the populations / Observed Data in the samples)

Big Issue:

In Classic statistics*, it is meaningless to talk about the Probability of Hypotheses*:
Hypotheses are either True or False.

Remark: see Bayesian stats for another view.

Hypothesis testing

In this framework, one computes:

p-value = P(Data is as extreme or more than Observed Data / H0=no difference between the populations)

Then compares it to an a priori statistical threshold, i.e., α = 5%. If p-value< α; then one rejects the H0.

The rationale: this approach ensures that you will falsely reject H0 (=False Alarm by stating that there is a diffenrence) on average α = 5% of the times you use this approach.

German	French
182	168
171	176
178	157
189	175

Idea: if there is no difference between French and German populations, French & GErman are interchangeable. In other words, we can arbitrarily sort participants in the two groups.

Let do that, many times, and compute the height difference between the groups obtained each time.

Then draw the histogram of these differences, and locate the rank of the observed difference in the real groups.

```
group1 = np.array([182, 171, 178, 189])
group2 = np.array([168, 176, 157, 175])

npermutations=10000
n1 = len(group1)
n2 = len(group2)
obs_diff = np.mean(group1) - np.mean(group2)

all = np.hstack((group1, group2))
diffs = []
for _ in range(npermutations):
        random.shuffle(all)
        list1 = all[:n1]
        list2 = all[n1:]
        diffs.append(np.mean(list1) - np.mean(list2))
sns.histplot(diffs)
np.mean(obs_diff >= diffs)
```

Estimation: Confidence intervals

Estimate the mean of a distribution (e.g; the height of French people), by taking a sample of N persons.

```
a = rnorm(100000, mean=100, sd=10)

N = 10
estimates = replicate(1000, mean(sample(a, N)))
sd(estimates)
```

```
import numpy as np

a = np.random.normal(loc=100, scale=10,
size=100000)

N = 10
estimates = np.array(
[np.mean(np.random.choice(a, size=N)) for _ in
range(1000)] )

np.std(estimates)
```

1. Do a full installation of goxpyriment (Go, Git + clone goxpyriment);

https://chrplr.github.io/goxpyriment/Installation/#full-installation

2. Open a terminal and run

cd goxpyriment
go run ./examples/choice-reaction-times/

3. Analyse the data (in the csv file)